Copy-on-Write Memory Demonstrator

Screenshots of Execution

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Part 1: C program (terminal, Ubuntu/WSL)

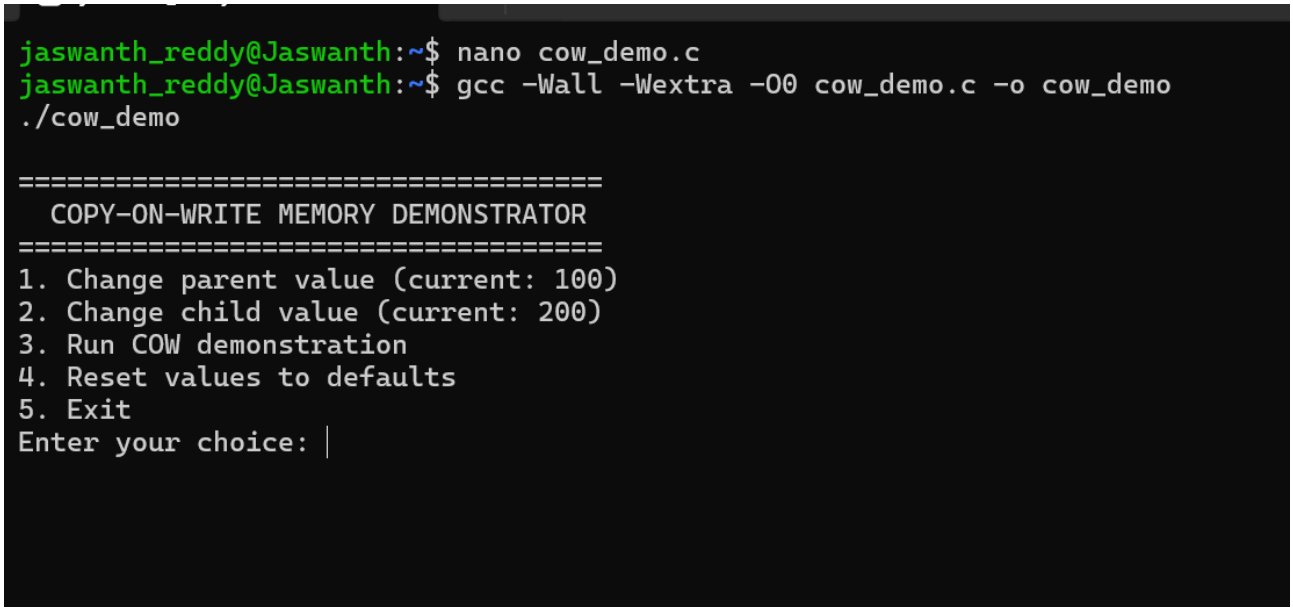


Figure 1. Compiling cow_demo.c with GCC and starting the menu (parent value 100, child value 200).

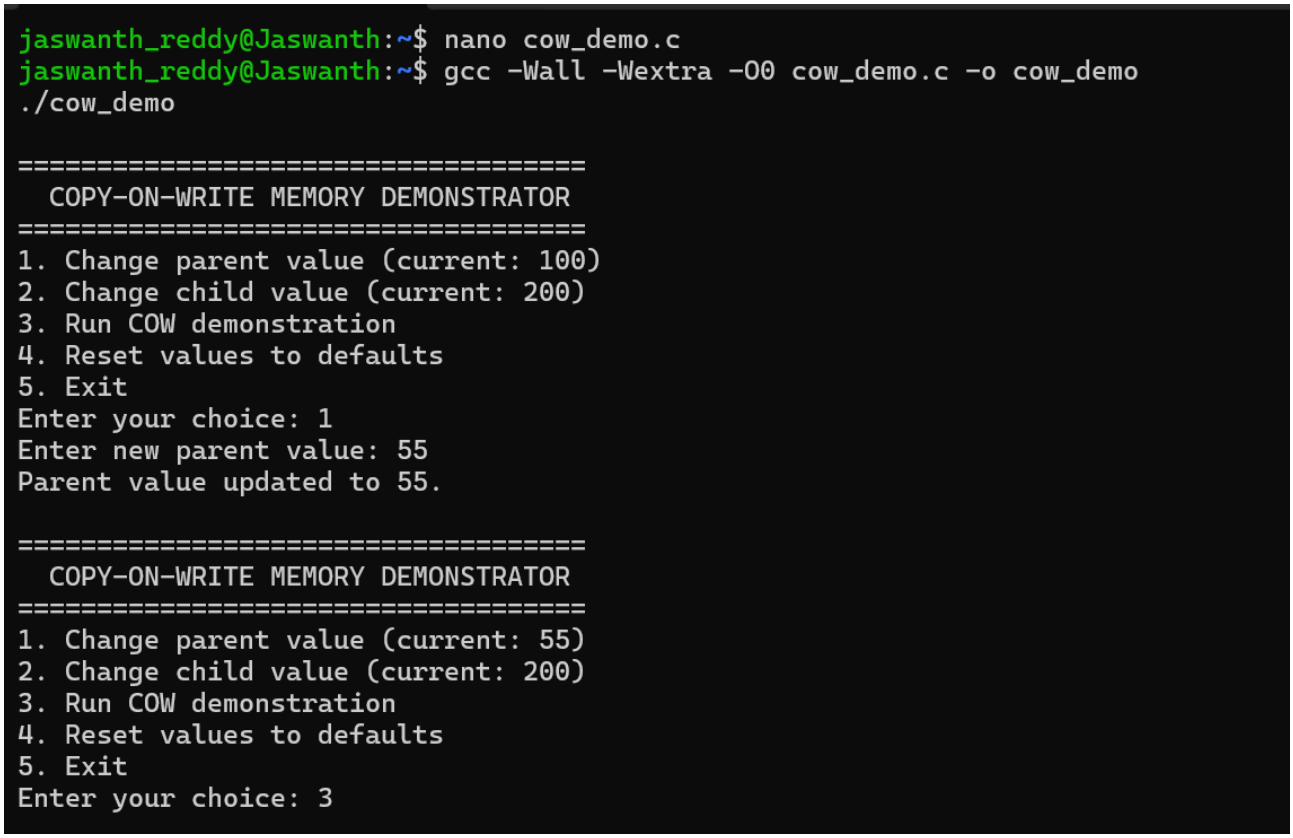


Figure 2. Changing the parent value to 55 (option 1), then selecting option 3 to run the COW demonstration.

```
========== COW DEMONSTRATION ==========
Initial value: 55
Parent PID: 1263

Child PID: 1271
Child reads initial value: 55
Child changes value to: 200
Child's final value: 200

Parent's value after child exits: 55
RESULT: Parent value is unchanged.
Copy-on-Write behavior demonstrated.
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```

Figure 3. Demonstration output: the child changes the value from 55 to 200, while the parent still reads 55 after the child exits.

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Parent's value after child exits: 55
RESULT: Parent value is unchanged.
Copy-on-Write behavior demonstrated.
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  COPY-ON-WRITE MEMORY DEMONSTRATOR
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1. Change parent value (current: 55)
2. Change child value (current: 200)
3. Run COW demonstration
4. Reset values to defaults
5. Exit
Enter your choice:
```

Figure 4. Result message ("Parent value is unchanged") and return to the menu.

Part 2: Web dashboard

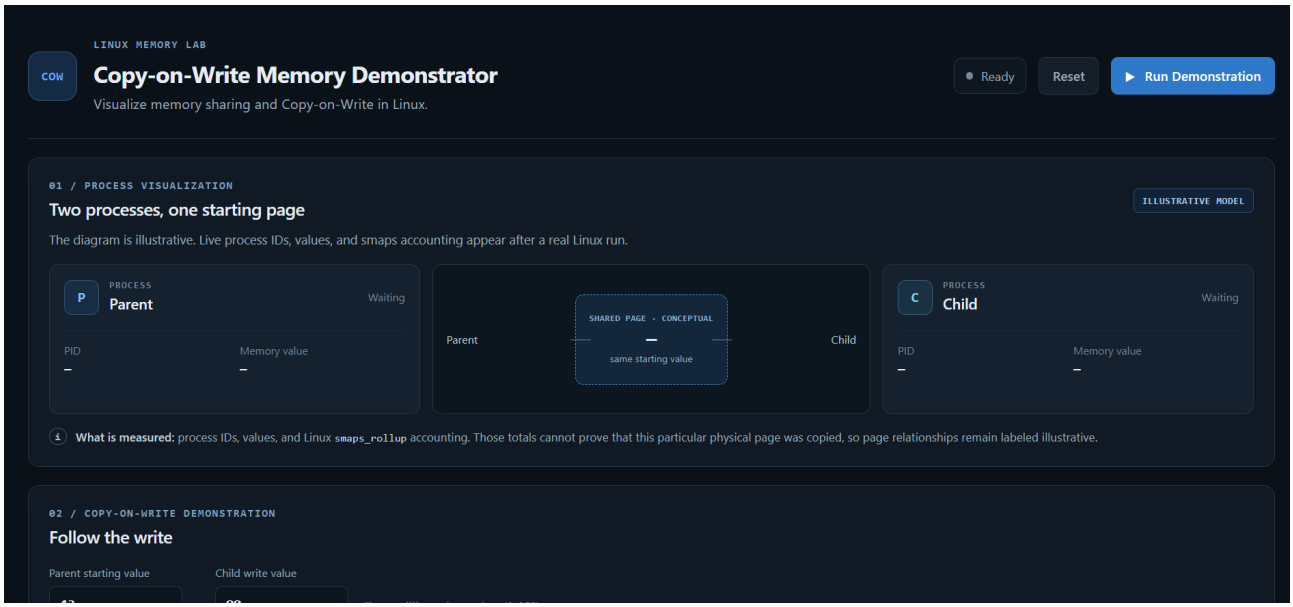


Figure 5. Dashboard before a run: process visualization marked as illustrative, parents/child waiting.

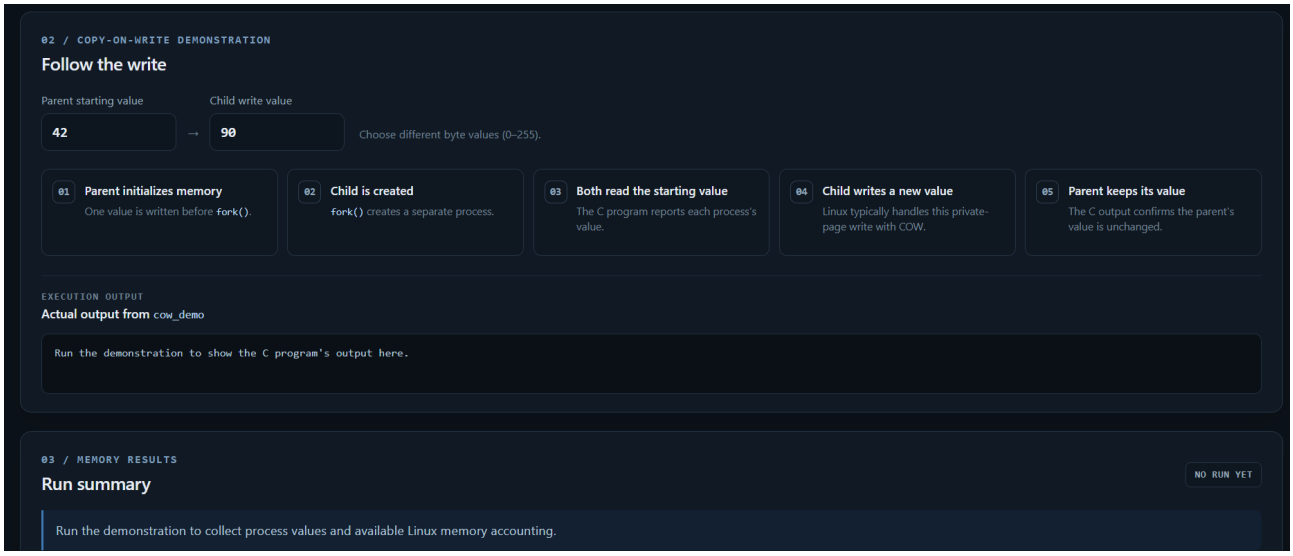


Figure 6. Follow-the-write section with parent value 42 and child value 90, before running.

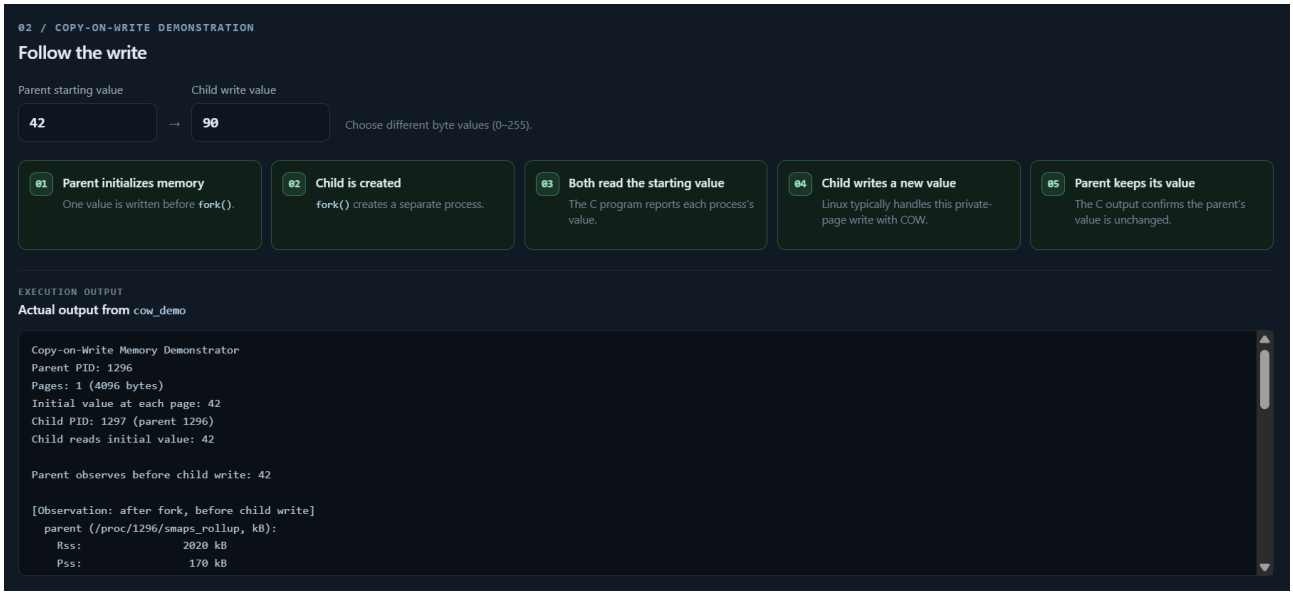


Figure 7. Actual output from cow_demo (PIDs 1296 and 1297) with /proc smaps_rollup readings.

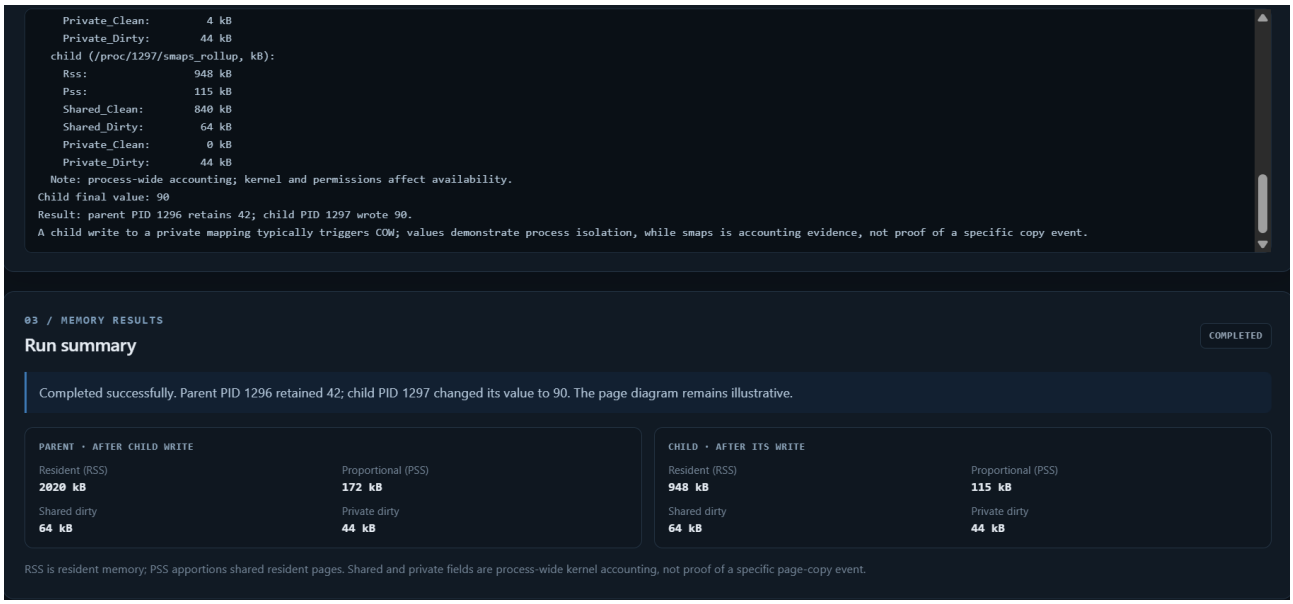


Figure 8. Run summary: parent PID 1296 kept 42, child PID 1297 changed its value to 90.